LABORATORY 3

HIGH-SPEED LATCH AND FLIP-FLOP DESIGN

OBJECTIVES

No.	Objectives	Requirements
1	Analyze the operation of DFF	▪ Explain the behavior and function at each internal node in the DFF
2	Measure the timing performance of DFF	▪ Understanding the concept of setup time and hold time ▪ Measure slew, CK to Q, setup time and hold time of an DFF
3	Optimize the setup and hold time intrinsic of DFF	▪ Verify sizes of DFF to meet the intrinsic setup time and hold time requirements.
4	Design a counter and calculate the timing margin	▪ Design a counter with given state machine. ▪ Optimizing and calculating the timing margin of this counter.

LAB 3 INFORMATION

- Consider ***NMOS nmos1v_lvt device, PMOS pmos1v_lvt device*** in ***gpdk45nm*** and power supply $vdd = 1.0V$
- **Note:** All figures in the document are for reference only.

EXPERIMENT 1

1.1. Objective: Analyze the operation of D Flip-flop (DFF).

1.2. Requirements: Simulate transient analysis with ADE-Explorer environment.

1.3. Instruction: Assemble the circuit as shown in ***Figure 2*** and ***Figure 3***, setting power supply parameters by **vdc** source, $vdd = 1.0V$, and the capacitor load $C_{load} = 1fF$. The transistor lengths $L_n = L_p = 45nm$, and W_N, W_P should be configured according to **Table 1.**

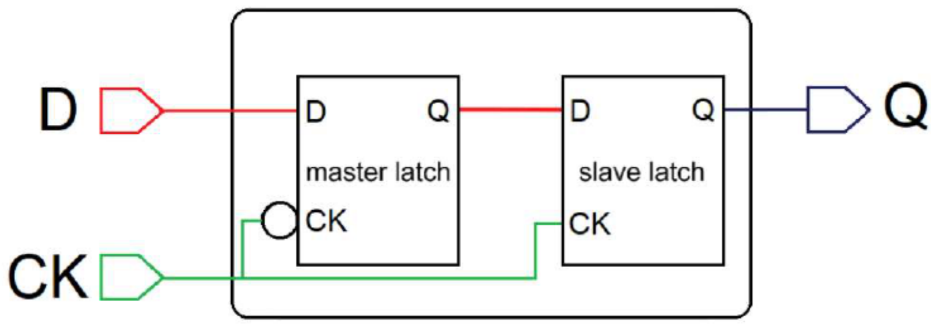


Figure 1. D Flip-flop diagram

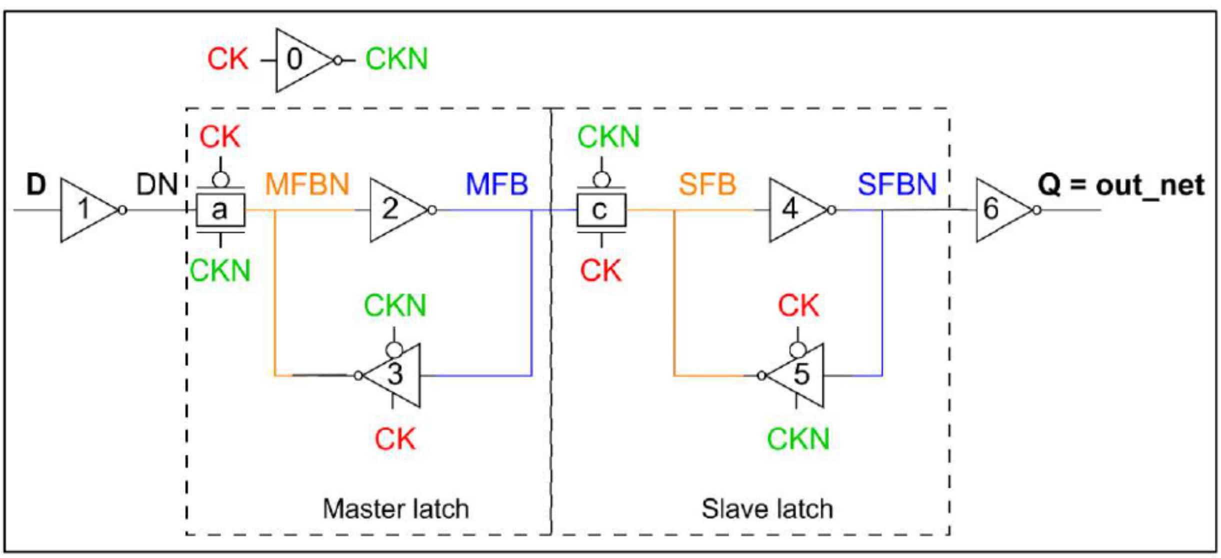


Figure 2. Architecture of a D flip-flop based on master–slave latch configuration

- ***Figure 3*** depicts the architectures of the inverter, inverter-Z, and transmission gate. Students use these three device types, with sizes specified in ***Table 1***, to implement a D flip-flop in Virtuoso.

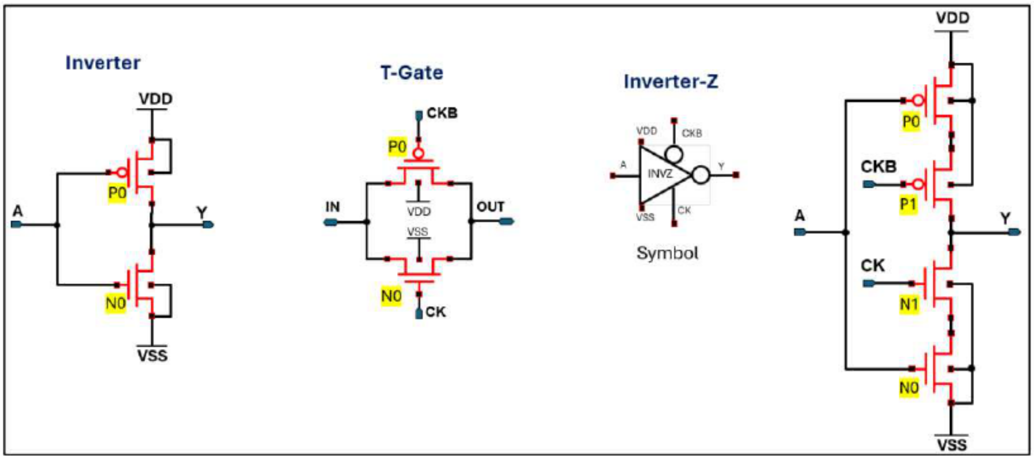


Figure 3. The architecture of INV, INV-Z and TGATE

Device sizes:

Device name	Device type	Length	Width PMOS	Width NMOS
0, 1	Inverter	45nm	P0: 120nm	N0: 120nm
a, c	T-Gate	45nm	P0: 120nm	N0: 120nm
2, 4, 6	Inverter	45nm	P0: 240nm	N0: 240nm
3, 5	Inverter-Z	45nm	P0: 240nm P1: 120nm	N0: 240nm N1: 120nm

Table 1. References size of devices in DFF

- Assemble the testbench as shown in ***Figure 4*** in Virtuoso, and set the *vpulse* source parameters (Voltage 1, Voltage 2, period, delay, rise time, fall time, and pulse width) for the **DATA** and **CLOCK** signals as shown in ***Figure 5***.

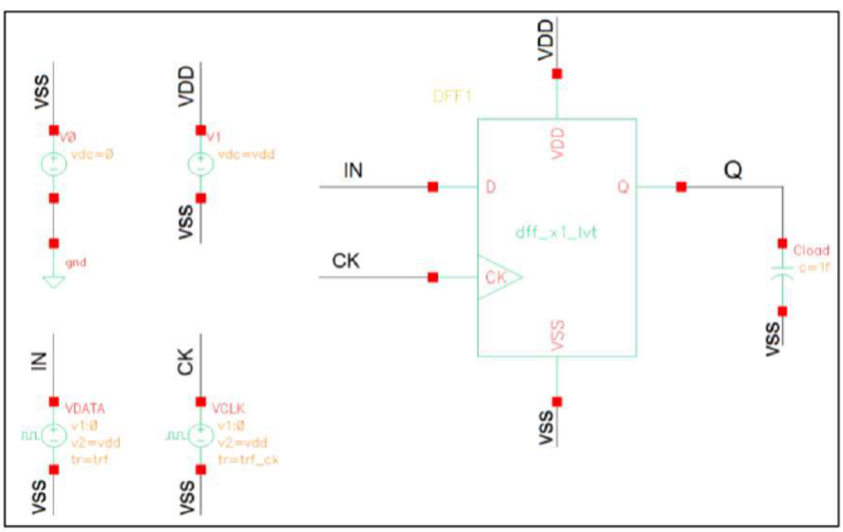


Figure 4. Testbench tran analysis of DFF

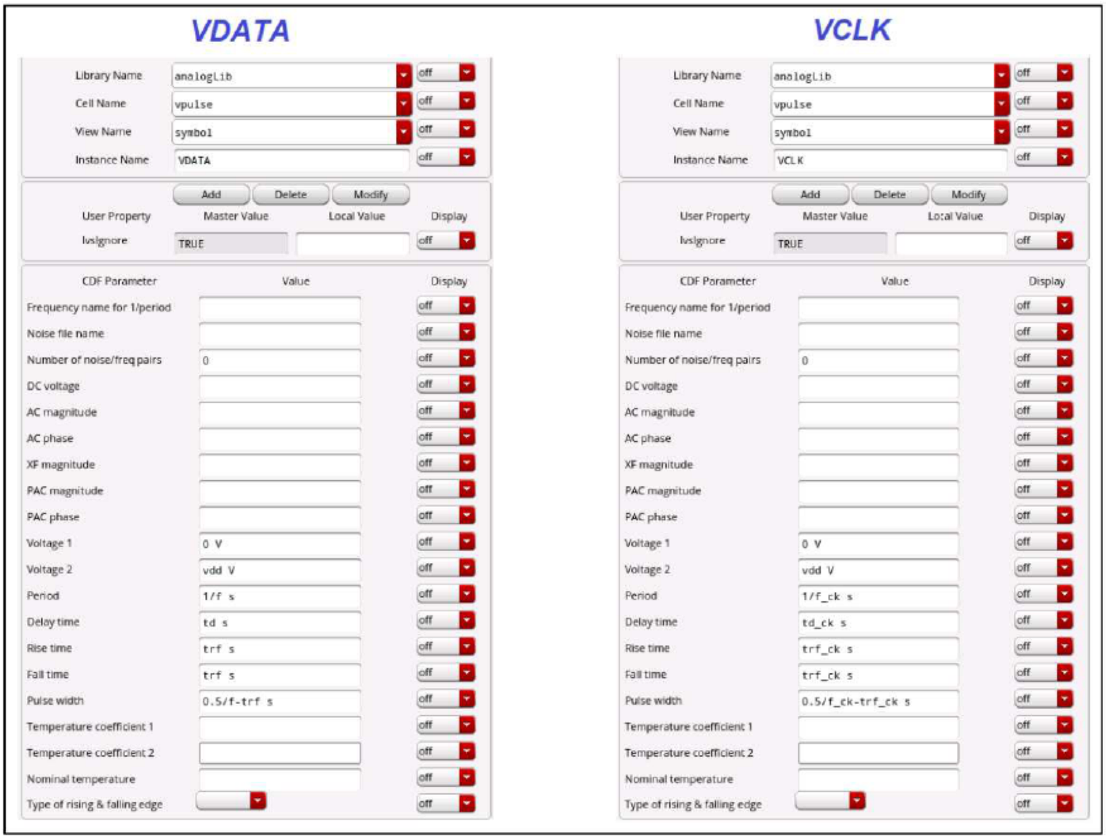


Figure 5. Testbench tran analysis of DFF

1.4. Analyze and report:

➢ *1.4.1. Testcase:*

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	*25°C*
$\boldsymbol{V_{DD}}$	$1\ V$
Data frequency	2G
Data slew	0.08*T_data
CK frequency	4G
CK slew	0.08*T_CK
Skew between Data and CK	0.5*T_CK
Observe	***Output and internal nodes of DFF***

- Students may refer the setup parameters shown in ***Figure 6***.

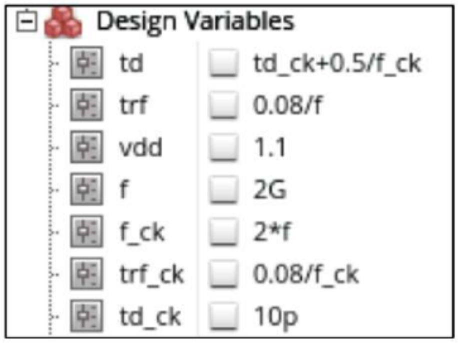


Figure 6. Setup parameters for transient analyses

- **Please provide the schematic and testbench here.**

➢ *1.4.2. Waveform:*

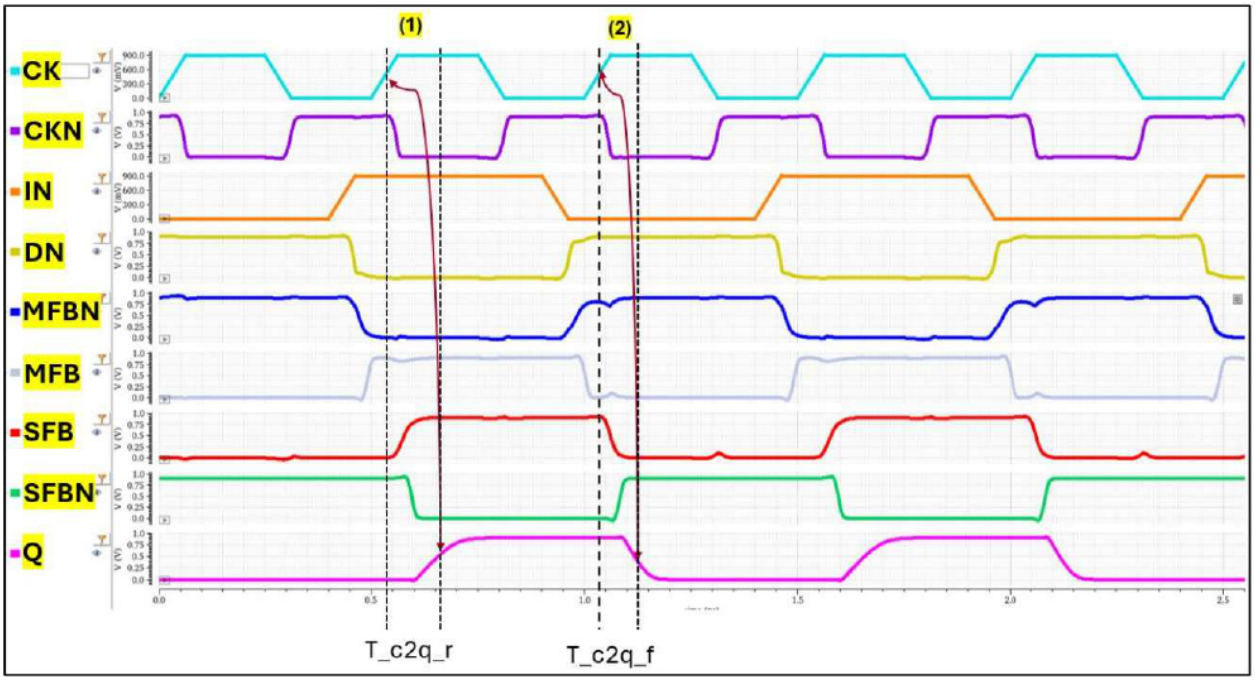


Figure 7. Output waveform of DFF

➢ *1.4.3. Observe from waveform:*

Complete the logic at each node for every state of the D flip-flop based on the previous waveform

State	CK	D	DN	tgate a	MFBN	invz 3	MFB	tgate c	SFB	SFBN	Q
Initial	**-**	**-**	**-**	**-**	**0**	**-**	**1**	**-**	**1**	**0**	**1**
(1)	**0**	**0**	0	on	0	on	0	on	0	0	0
			1	off	1	off	1	off	1	1	1

(2)	**0->1**	**0**	0 1	on->off off->on	0 1	on->off off->on	0 1	on->off off->on	0 1	0 1	0 1
(3)	**1**	**0**	0 1	on off	0 1	on off	0 1	on off	0 1	0 1	0 1
(4)	**1**	**1**	0 1	on off	0 1	on off	0 1	on off	0 1	0 1	0 1
(5)	**1->0**	**1**	0 1	on->off off->on	0 1	on->off off->on	0 1	on->off off->on	0 1	0 1	0 1
(6)	**0**	**0**	0 1	on off	0 1	on off	0 1	on off	0 1	0 1	0 1
(7)	**0**	**1**	0 1	on off	0 1	on off	0 1	on off	0 1	0 1	0 1
(8)	**0->1**	**1**	0 1	on->off off->on	0 1	on->off off->on	0 1	on->off off->on	0 1	0 1	0 1

- **Complete the truth table of D-Flip Flop based on the previous waveform**

D	CK	Q
0	**0**	**Q*/0/1**
1	**0**	**Q*/0/1**
0	**1**	**Q*/0/1**
1	**1**	**Q*/0/1**
0	**1->0**	**Q*/0/1**
1	**1->0**	**Q*/0/1**
0	**0->1**	**Q*/0/1**
1	**0->1**	**Q*/0/1**
Q is the previous state of Q*		

➢ <u>*1.4.4. Explanation (if applicable):*</u>

- ......

EXPERIMENT 2

2.1. *Objective:* Measure the transient response of D Flip-flop (DFF).

2.2. *Requirements:* Simulate transient analysis with ADE-Explorer environment.

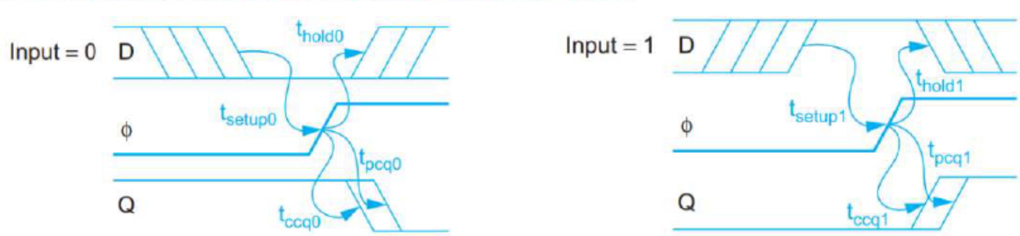
2.3. *Instruction:* Assemble the circuit as shown in ***Figure 2*** and ***Figure 3***, setting power supply parameters by ***vdc*** source, $vdd = 1.0V$, and the capacitor load $C_{load} = 1fF$. The transistor lengths $L_n = L_p = 45nm$, and W_N, W_P should be configured according to **Table 1.**

2.4. *Analyze and report:*

➢ *2.4.1. Testcase:*

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	**TT, SS, FF**
Temp	$-\mathbf{40}℃, \mathbf{25}℃, \mathbf{125}℃$
$\boldsymbol{V_{DD}}$	$\mathbf{0.9V, 1\,V, 1.1V}$
Data frequency	2G
Data slew	0.08*T_data
CK frequency	4G
CK slew	0.08*T_CK
Skew between Data and CK	0.5*T_CK
Observe	***Output and internal nodes of DFF*** ***Measure output slew, CK to Q, Setup and Hold time sim, Power***

- Assemble the testbench as shown in ***Figure 4*** in Virtuoso, and set the *vpulse* source parameters (Voltage 1, Voltage 2, period, delay, rise time, fall time, and pulse width) for the DATA and CLOCK signals as shown in ***Figure 5***.
- Students may refer the setup parameters shown in ***Figure 6***.
- **Please provide the schematic and testbench here.**



➢ *2.4.2. Waveform:*

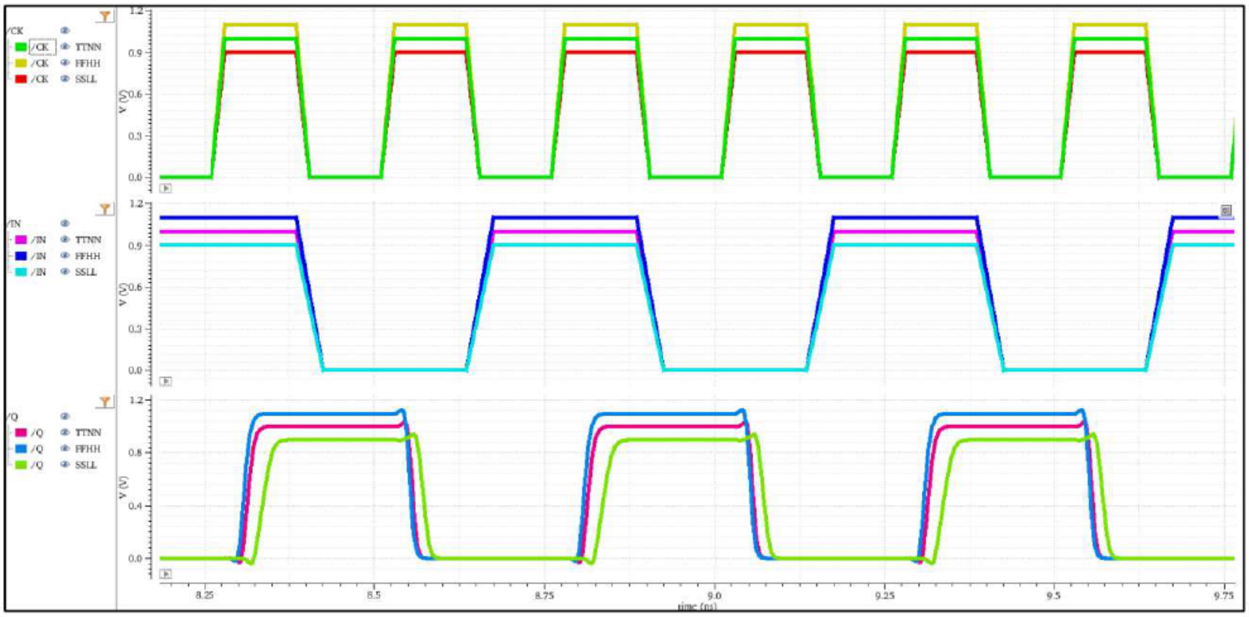


Figure 8. Output transient result of DFF

➢ *2.4.3. Observe from waveform:*

Corners	Setup time sim Data 0	Hold time sim Data 0	Setup time sim Data 1	Hold time sim Data 1	CK-to-Q rise delay	CK-to-Q fall delay	Trise-Q 10%-90%	Tfall-Q 10%-90%	Power
SSLL									
FFHH									
TTNN									

- **Include the output results after simulation, as shown in Figure 9, for reference and comparison.**

	Parameter						TTNN	FFHH	SSLL
	gpdk045.scs						tt	ff	ss
	temperature						25	125	-40
	vdd						1	1.1	900m

13 rows

Test	Output	Spec	Weight	Pass/Fail	Min	Max	TTNN	FFHH	SSLL
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
nttrung_tb_tb_d...	t_rise				17.11p	24.46p	17.72p	17.11p	24.46p
nttrung_tb_tb_d...	t_fall				12.64p	16.43p	12.64p	13.09p	16.43p
nttrung_tb_tb_d...	tr_c2q				2.539n	2.567n	2.545n	2.539n	2.567n
nttrung_tb_tb_d...	tf_c2q				2.658n	2.677n	2.661n	2.658n	2.677n

Figure 9. Output transient measurement result of DFF

➢ *2.4.4. Explanation (if applicable):*

- ……

EXPERIMENT 3

3.1. Objective: Characterize the intrinsic setup time and hold time of D Flip-flop (DFF).

3.2. Requirements: Simulate transient analysis with ADE-Explorer environment.

3.3. Instruction: Assemble the circuit as shown in ***Figure 2, Figure 3*** and ***Figure 4***, setting power supply parameters by **vdc** source, $vdd = 1.0V$, and the capacitor load $C_{load} = 1fF$. The transistor lengths $L_n = L_p = 45nm$, and W_N, W_P should be configured according to **Table 1.**

3.4. Analyze and report:

- *3.4.1. Testcase:*

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	**SS, FF**
Temp	$\mathbf{-40}$℃, $\mathbf{125}$℃
$\boldsymbol{V_{DD}}$	$\mathbf{0.9V, 1.1V}$
Data frequency	2G
Data slew	0.08*T_data
CK frequency	4G
CK slew	0.08*T_CK
Observe	***Characterize intrinsic setup time and hold time of DFF***

- Assemble the testbench as shown in ***Figure 4*** in Virtuoso, and set the *vpulse* source parameters (Voltage 1, Voltage 2, period, delay, rise time, fall time, and pulse width) for the DATA and CLOCK signals as shown in ***Figure 5***.
- Students may refer the setup parameters shown in ***Figure 10***.

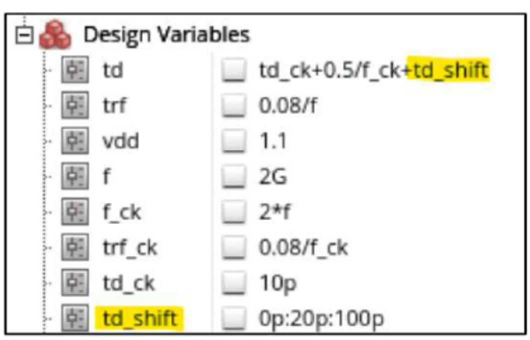


Figure 10. Setup parameters to measure intrinsic setup/hold time

- **Students should first perform a coarse sweep to determine the appropriate range, followed by a finer sweep to improve the accuracy of the results.** (vary ***td_shift*** variables)
- **Please provide the schematic and testbench here.**
- To characterize the intrinsic setup and hold times of the flip-flop, shift the DATA signal relative to the CLK edge until the flip-flop fails to capture the input correctly. The boundary between pass and fail conditions is defined as the target value. Shift the DATA signal to the right to measure the setup time and to the left to measure the hold time.

➢ *3.4.2. Measure intrinsic setup time:*

- ***Figure 11*** illustrates the waveform and method used to measure the intrinsic setup time for data '1' of the D flip-flop. The delay between the data input (IN) and clock (CK) is reduced from 32.5ps to 20ps. When the delay is 30ps, the output transitions to a high level, indicating that the DFF correctly captures the input data. However, at a delay of 27.5ps, the output remains at a low logic level, showing that the DFF fails to capture data '1'. Therefore, the intrinsic setup time for data '1' is determined to be approximately 28.75 ps.

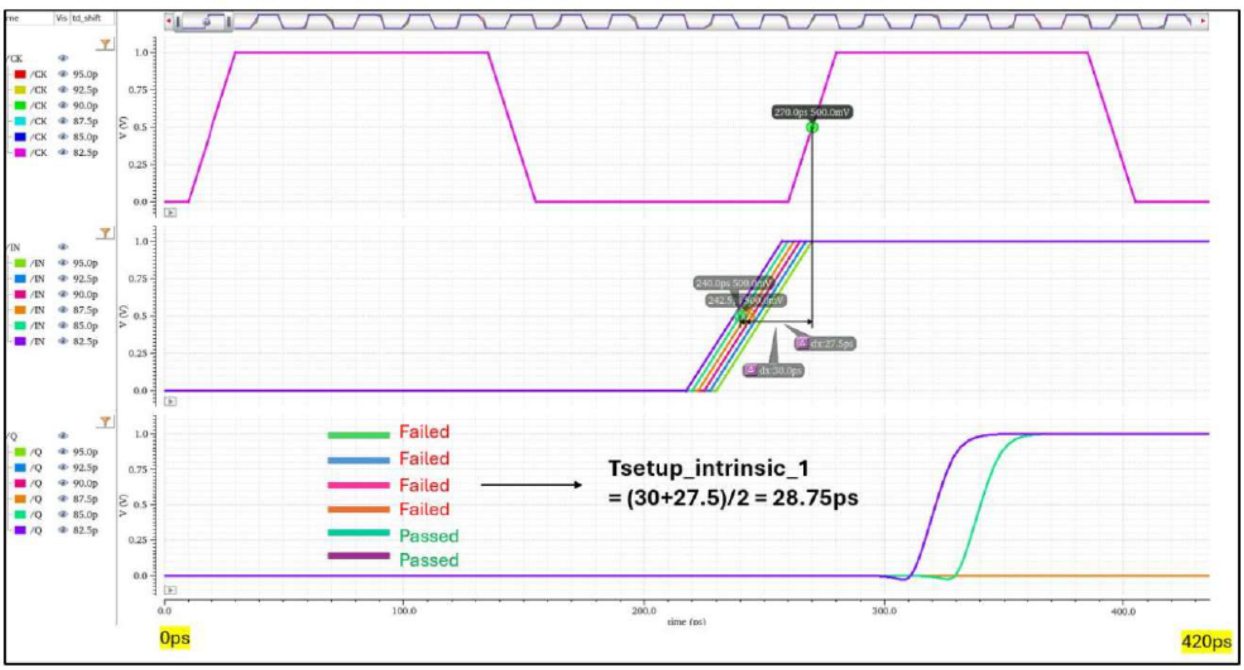


Figure 11. The waveform of setup time intrinsic data 1 of DFF

Students should use the first two pulses (from 0ns to 450ns) to measure the intrinsic setup and hold times, as the correct state during these pulses is known. Using subsequent

pulses may lead to inaccurate results because the correctness of the previously captured state is uncertain.

- Students should apply the same method to measure the intrinsic setup time for data '0'. Configure the DATA signal to start at logic '1' instead of logic '0', as shown in ***Figure 12*** (*Hint: modify the VDATA source*).

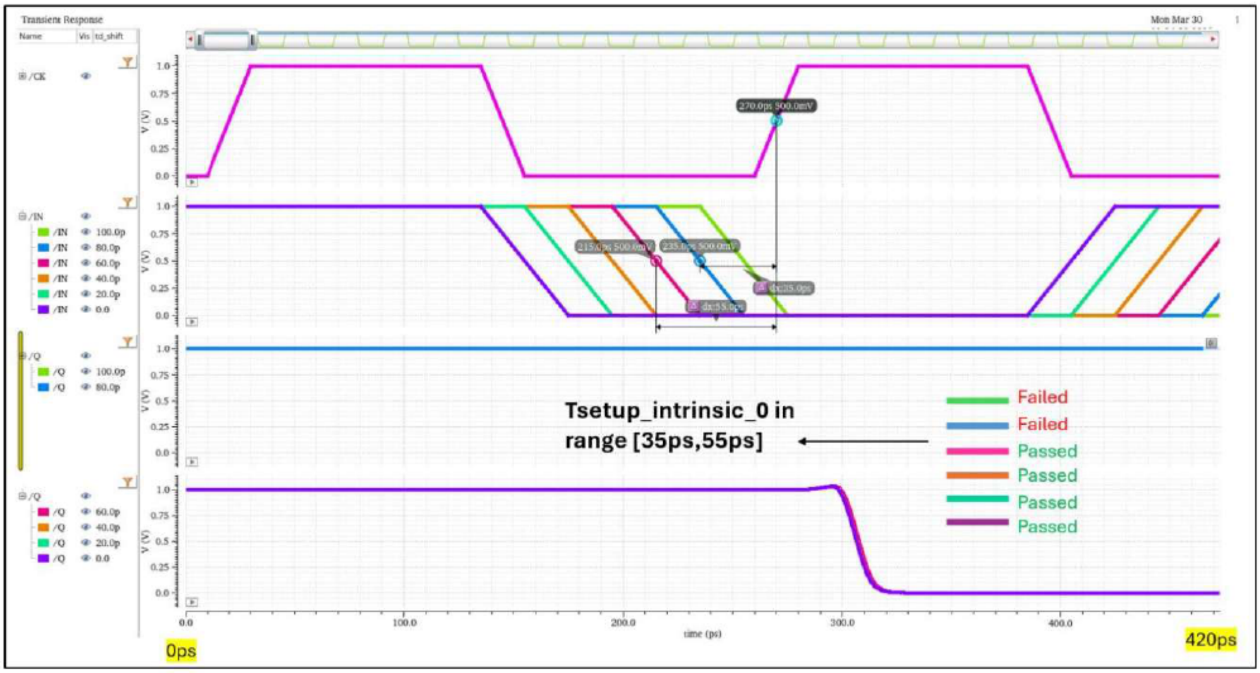


Figure 12. The waveform of setup time intrinsic data 0 of DFF

- **Report 4 waveforms used to measure the intrinsic setup time, as shown in Figure 11 and Figure 12.**
- **Collect the intrinsic setup time results at FFHH and SSLL corners.**

Corner	Intrinsic setup data 0	Intrinsic setup data 1
FFHH		
SSLL		

- **Explain the origin of the intrinsic setup time requirement in a D flip-flop, with reference to the master–slave latch architecture and the associated timing waveforms.** (*Explain the response at each internal node of DFF as* ***Figure 13***)

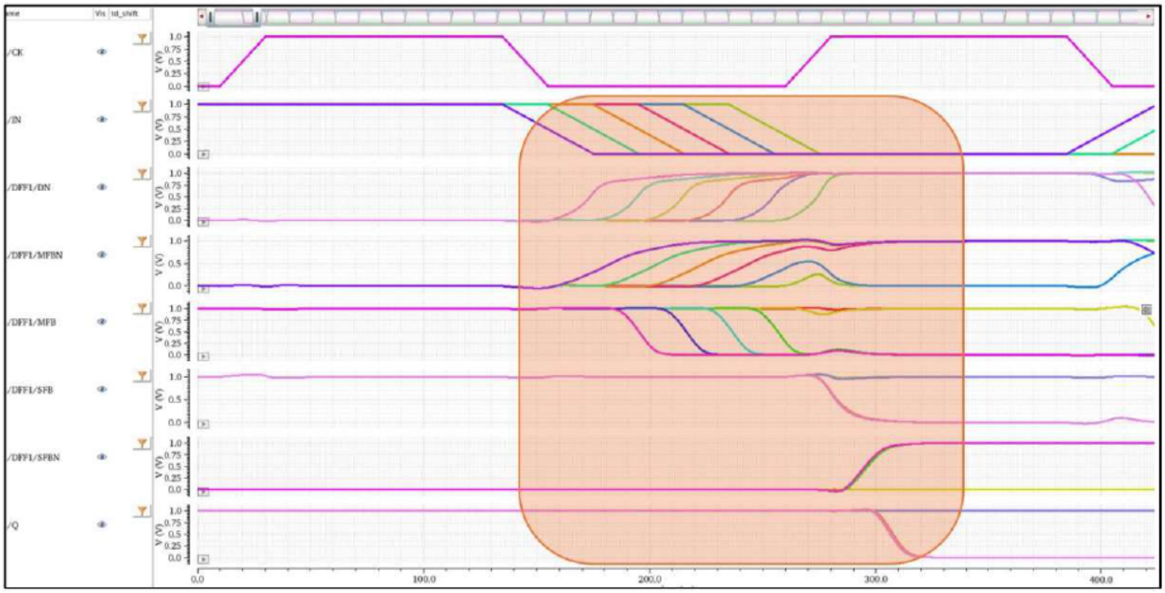


Figure 13. The waveform internal nodes of DFF

- *3.4.3. Measure intrinsic hold time:*

- Change the VDATA source from ***vpulse*** to ***vpwl*** and configure parameters as shown in ***Figure 14***.

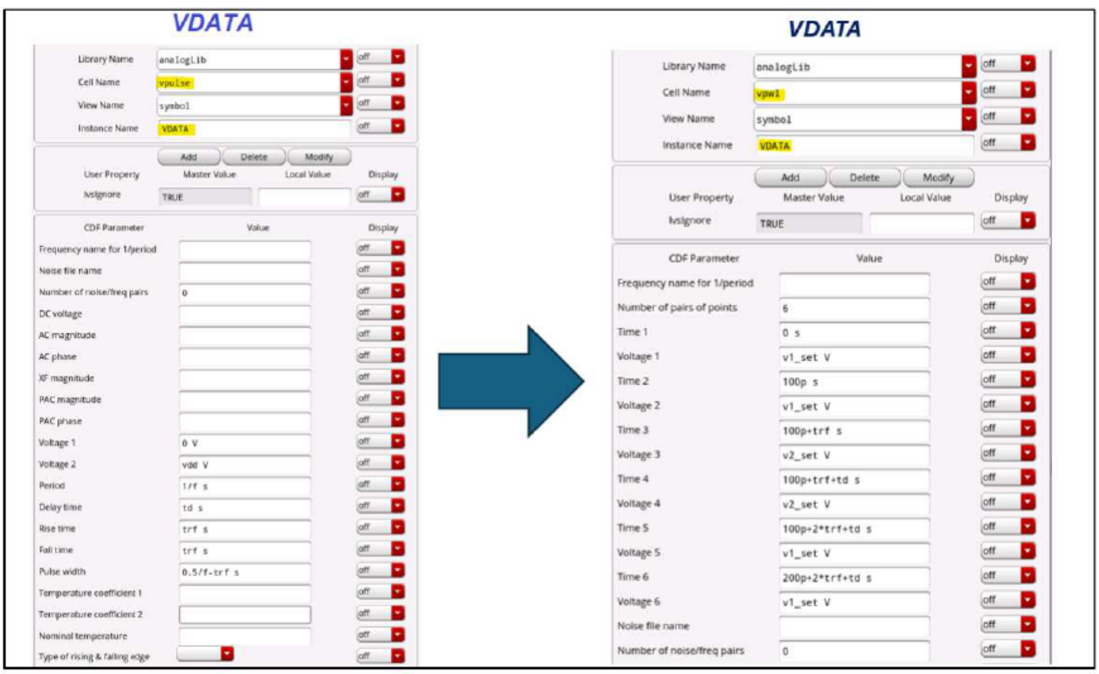


Figure 14. Change VDATA source from vpulse to vpwl

- Set up variables as shown in ***Figure 15*** and vary ***td_shift, v1_set*** and ***v2_set*** to measure the intrinsic hold time

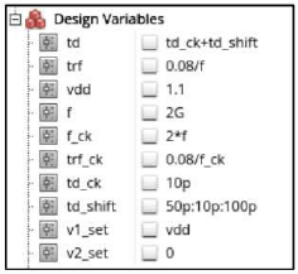


Figure 15. Setup parameters for intrinsic hold time measuring

- Similar to the measurement of intrinsic setup time, the intrinsic hold time is determined from the waveforms shown in **Figures 16** and **17.**

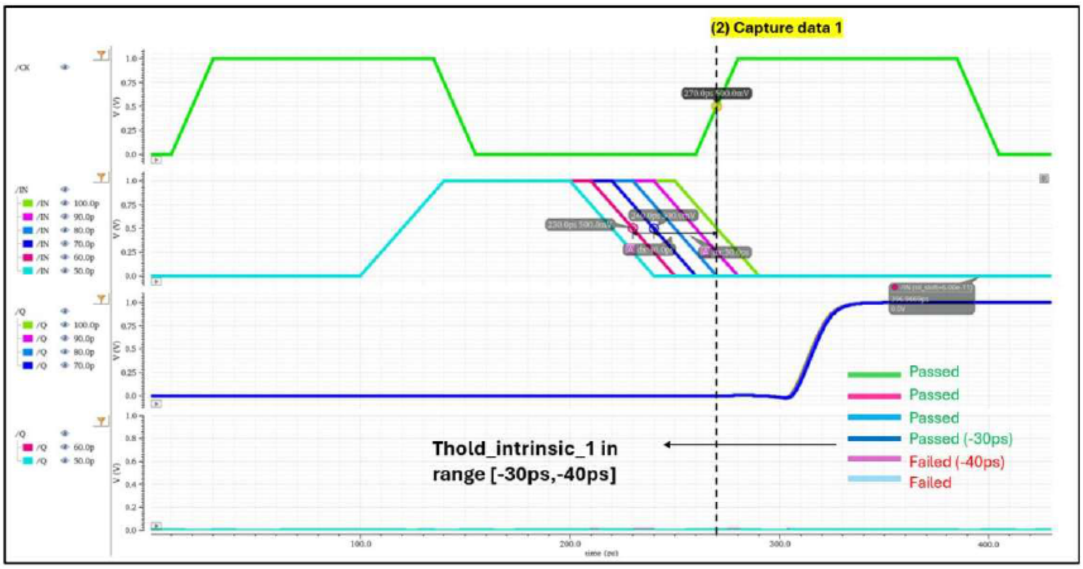


Figure 16. The waveform of hold time intrinsic data 1 of DFF

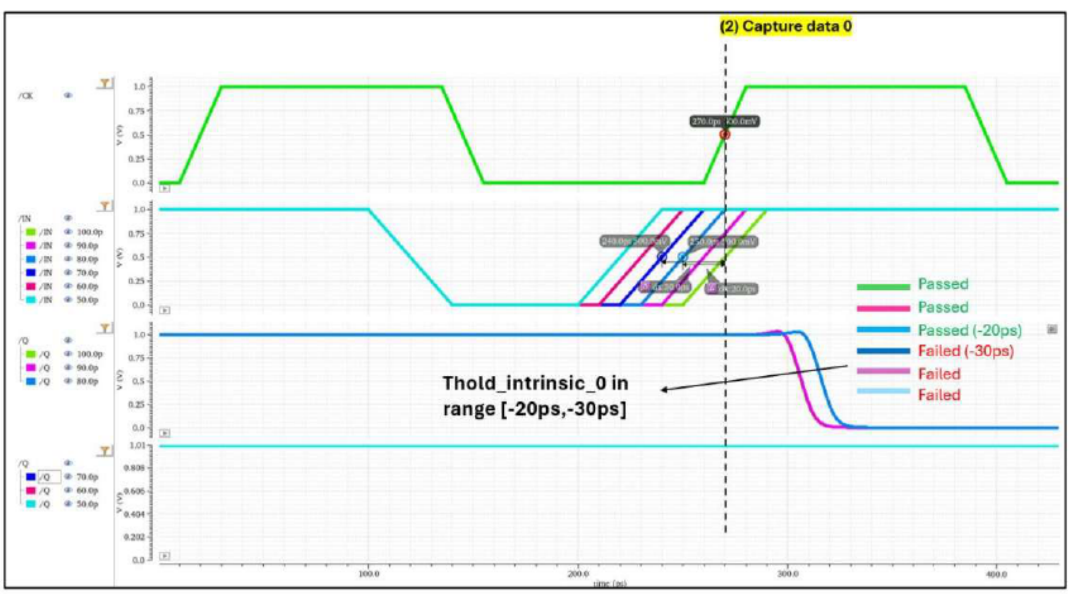


Figure 17. The waveform of hold time intrinsic data 0 of DFF

- **Report 4 waveforms used to measure the intrinsic hold time, as shown in Figure 16 and Figure 17.**
- **Collect the intrinsic hold time results at FFHH and SSLL corners.**

Corner	Intrinsic hold data 0	Intrinsic hold data 1
FFHH		
SSLL		

- **Explain why the intrinsic hold time of a D flip-flop can be negative, based on the master-slave latch architecture and the associated timing waveforms.** *In your explanation, analyze the behavior of each internal node of the DFF, as shown in* ***Figure 18****.*

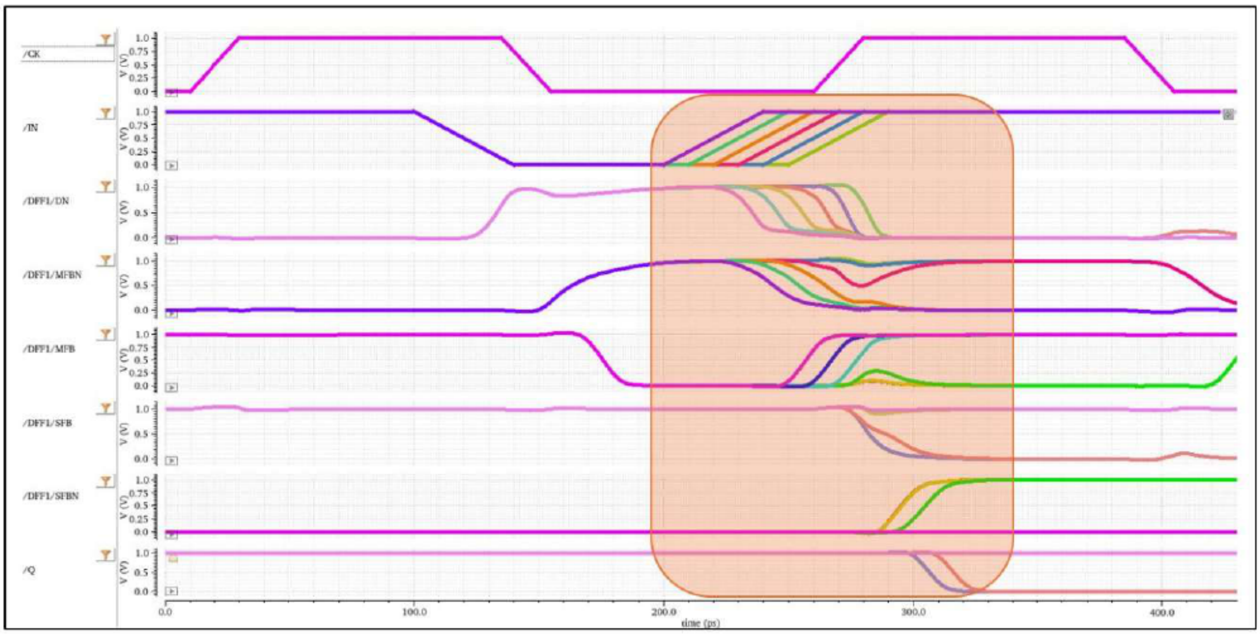


Figure 18. The waveform internal nodes of DFF

EXPERIMENT 4

4.1. Objective: Design a D Flip-flop (DFF) with target intrinsic setup time, hold time and slew.

4.2. Requirements: Simulate transient analysis with ADE-Explorer environment.

4.3. Instruction: Assemble the circuit as shown in ***Figure 2, Figure 3*** and ***Figure 4***, setting power supply parameters by **vdc** source, $vdd = 1.0V$, and the capacitor load $C_{load} = 1fF$. The transistor lengths $L_n = L_p = 45nm$, and **W_N, W_P should be verify to meet the specification.**

4.4. Analyze and report:

- *4.4.1. Testcase:*

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	**SS, TT, FF**
Temp	$-\mathbf{40}$℃, $\mathbf{25}$℃, $\mathbf{125}$℃
$\boldsymbol{V_{DD}}$	$\mathbf{0.9V, 1.0V, 1.1V}$
Data frequency	2G
Data slew	0.08*T_data
CK frequency	4G
CK slew	0.08*T_CK
Observe	***Measure intrinsic setup time and hold time of DFF*** ***Output transient response***

- Measure intrinsic setup time and hold time as ***Experiment 3***.
- **Verify size (W_N, W_P) of each device to meet the following specification.**

Item	Spec	Unit	Note
Intrinsic setup Time (data 0, 1)	<40	ps	Worst case of 3 PVT FFHH SSLL TTNN
Intrinsic hold time (data 0, 1)	<40	ps	
clk-to-Q rise delay	<90	ps	
clk-to-Q fall delay	<90	ps	
Trise-Q 10%-90%	<50	ps	
Tfall-Q 10%-90%	<50	ps	
Power TT	As low as possible	uA	

- **Table size of NMOS and PMOS each device after verification.**

- *4.4.2. Waveform:*

- **Provide all waveform measure intrinsic setup time and hold time at each PVT.**
- **Measure trise, tfall and delay in this waveform also**. ***(Using marker, M and D key shortcut)***

- *4.4.3. Observe from waveform:*

Corners	Intrinsic Setup Time Data 0	Intrinsic Hold time Data 0	Intrinsic Setup Time Data 1	Intrinsic Hold time Data 1	CK-to-Q rise delay	CK-to-Q fall delay	Trise-Q 10%-90%	Tfall-Q 10%-90%	Power
SSLL									
FFHH									
TTNN									

- *4.4.4. Explanation (if applicable):*

- ……

EXPERIMENT 5

5.1. Objective: Design a counter with DFF and calculate timing margin.

5.2. Requirements: Simulate transient analysis with ADE-Explorer environment.

5.3. Instruction: Implement the counter working at 4GHz, $vdd = 1.0V$, and the capacitor load $C_{load} = 1fF$. **Use the D-Flip flop from Experiment 4 to design this counter.**

- Assemble the circuit as shown in ***Figure 19*** and setup pwl source as ***Figure 20.***

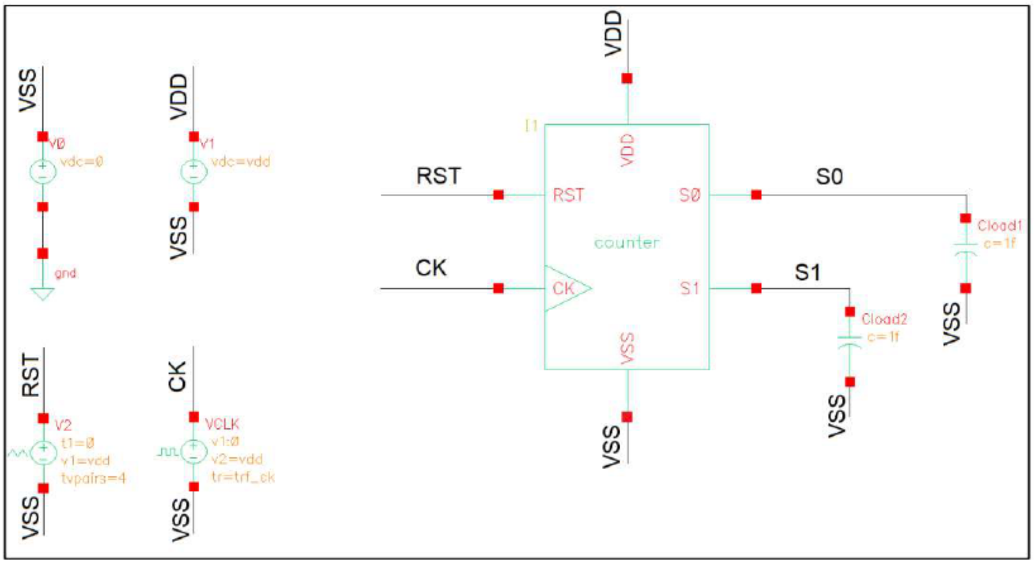


Figure 19. Testbench of Counter

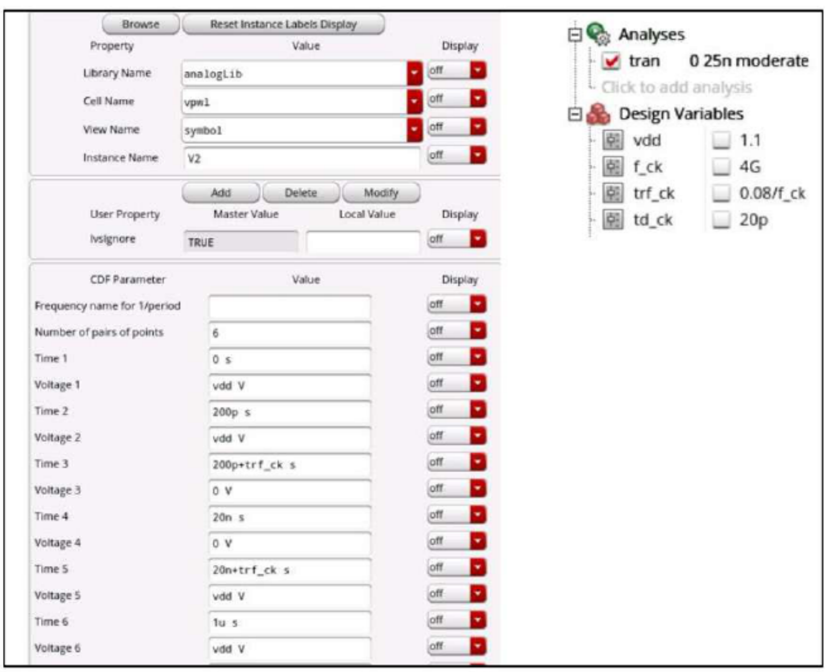


Figure 20. Setup parameter for RST source

5.4. Analyze and report:

S1	S0	S1+	S0+
0	0		
0	1		
1	0		
1	1		

- Analyze the state diagram and determine the state transitions. **Complete the state table.**
- Include a reset mechanism to initialize the counter to state S_1S_0=00.
- **Implement the design using D flip-flops and logic gates in Virtuoso.**
- **Please provide the schematic and testbench here.**

➢ *5.4.1. Testcase:*

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	**SS, TT, FF**
Temp	$-\mathbf{40}℃, \mathbf{25}℃, \mathbf{125}℃$
$\boldsymbol{V_{DD}}$	$\mathbf{0.9}\boldsymbol{V}, \mathbf{1.0}\boldsymbol{V}, \mathbf{1.1}\boldsymbol{V}$
Data frequency	2G
Data slew	0.08*T_data
CK frequency	4G
CK slew	0.08*T_CK
Observe	***Output transient response*** ***Measure setup and hold time available***

- **Please provide the schematic and testbench here.**

➢ 5.4.2. Waveform

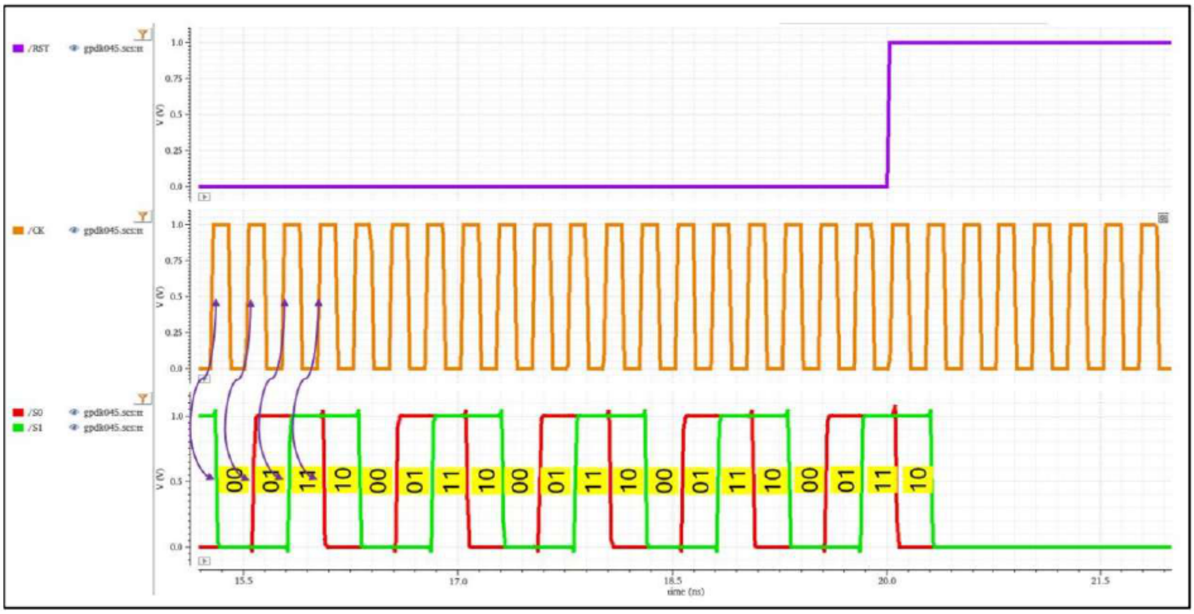


Figure 21. Output waveform of counter

- The waveform shows the corrected function at both RST = 0 and RST = 1.

➢ 5.4.3. Observe from waveform:

- Students should write the timing equations to calculate the setup time sim and hold time sim of each flip-flop. **Pay careful attention to the clock lunch and clock capture paths for FF_S0 and FF_S1. Please refer the detail in Figures 22, 23 and 24.**

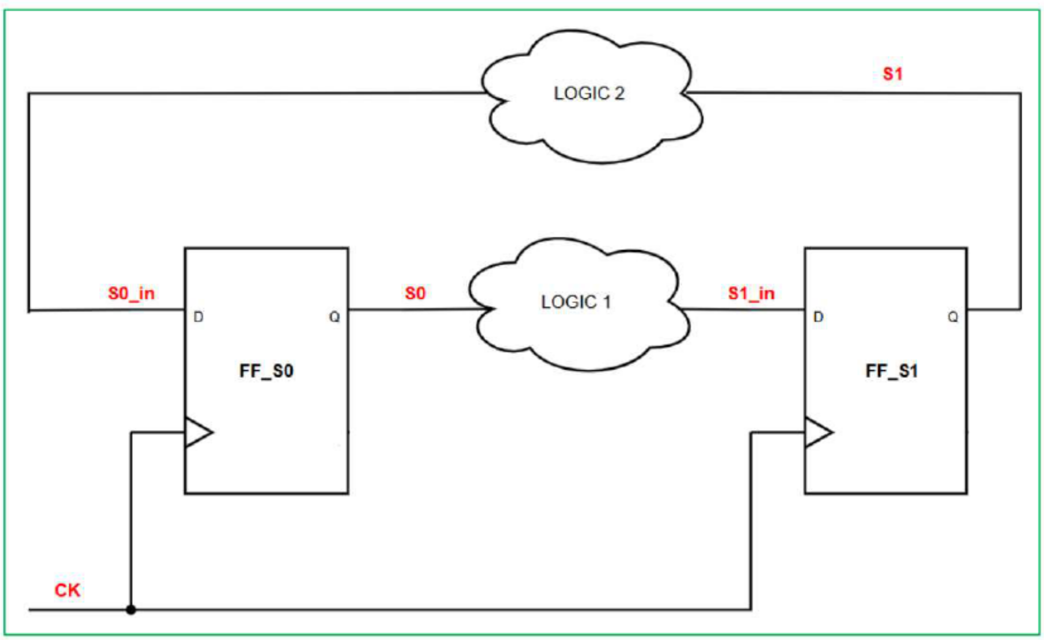


Figure 22. The diagram of FF_S0 and FF_S1

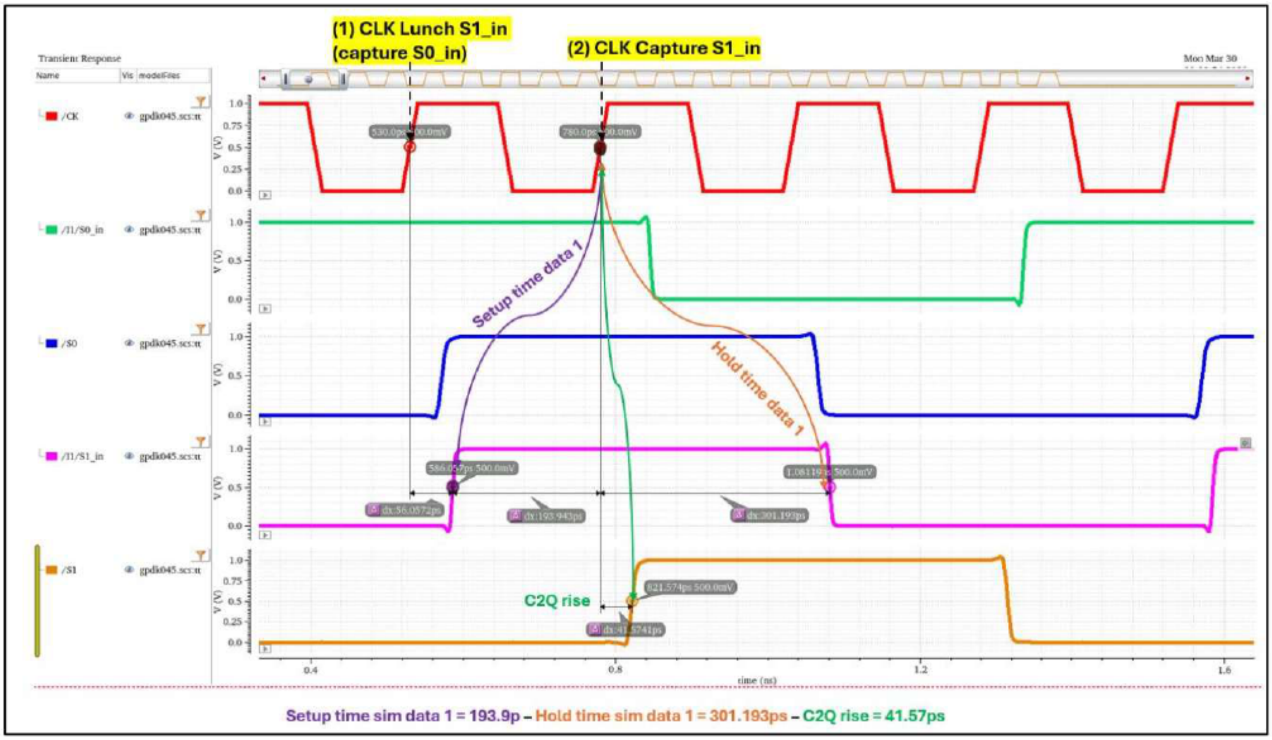


Figure 23. Example waveform for CLK Lunch and CLK Capture data 1 at FF_S1

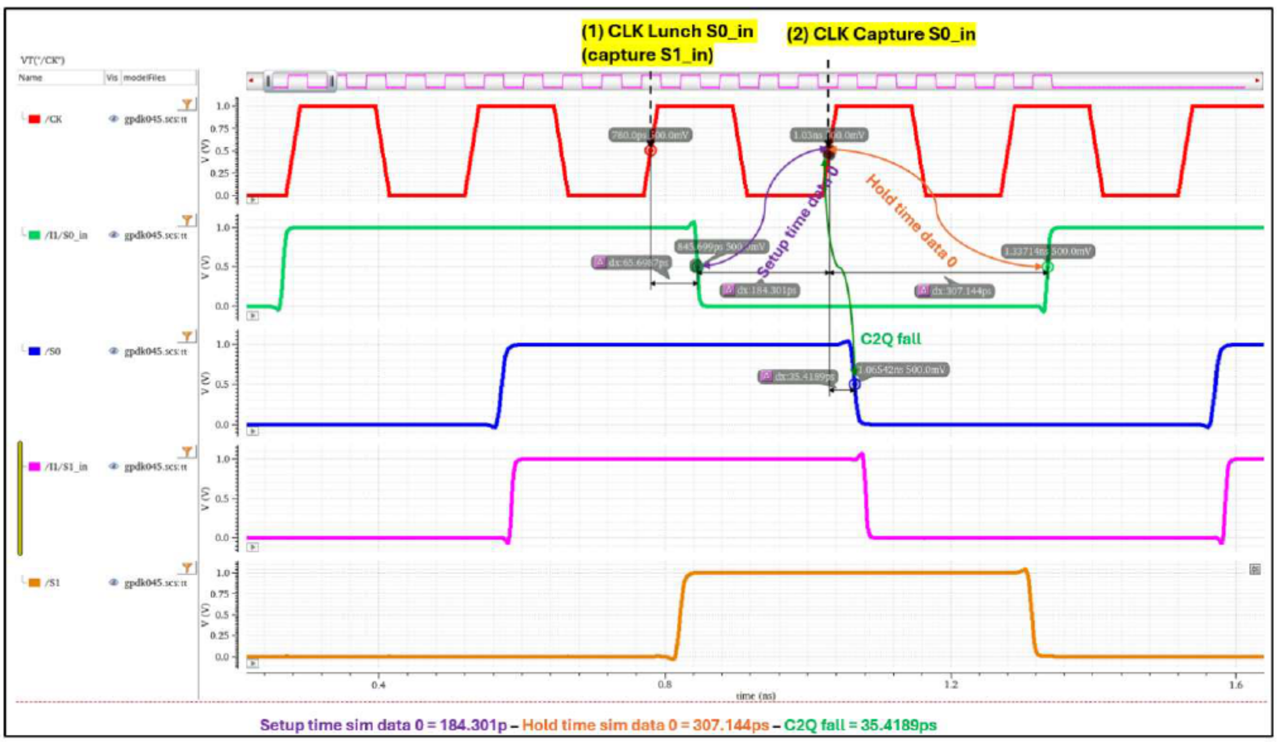


Figure 24. Example waveform for CLK Lunch and CLK Capture data 0 at FF_S0

- **Flip flop for S0:**

Corners	Setup time Sim Data 0	Hold time Sim Data 0	Setup time Sim Data 1	Hold time Sim Data 1	CK-to-S0 rise delay	CK-to-S0 fall delay	Trise-S0 10%-90%	Tfall-S0 10%-90%	Power
SSLL									
FFHH									
TTNN									

- **Flip flop for S1:**

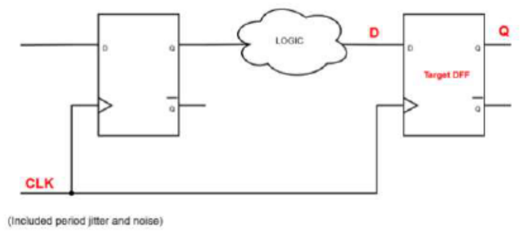
Corners	Setup time Sim Data 0	Hold time Sim Data 0	Setup time Sim Data 1	Hold time Sim Data 1	CK-to-S0 rise delay	CK-to-S0 fall delay	Trise-S0 10%-90%	Tfall-S0 10%-90%	Power
SSLL									
FFHH									
TTNN									

➢ *5.4.4. Explanation (if applicable):*

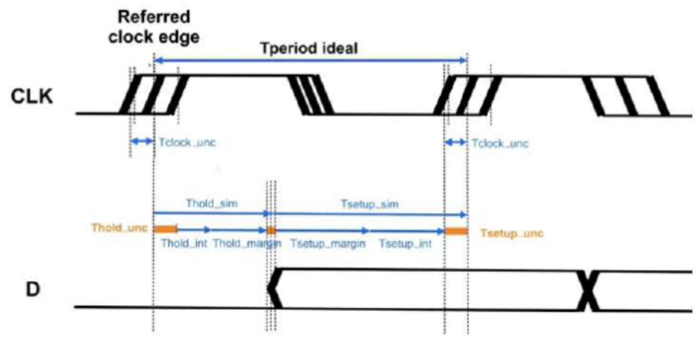
- ……

5.5. Calculate timing margin

- To ensure that all flip-flops operate reliably under variations in environmental conditions (e.g., Noise, Heating, Jitter, Monte Carlo variations…), the setup and hold times for each timing path (e.g., flip-flop to flip-flop, latch to flip-flop) must have sufficient timing margins.
- ***Figure 25*** depicts a proposed method for calculating the timing margin.
- The intrinsic setup time ($T_{setup,int}$) and intrinsic hold time ($T_{hold,int}$) are characterized in ***Experiment 4***, while the simulated setup time ($T_{setup,sim}$) and hold time ($T_{hold,sim}$) are obtained in ***Section 5.4***.
- In this lab, we assume $\boldsymbol{T_{clock\_unc}} = 4\% * T_{cycle\_ideal} = 4\% * 250ps \; = \; \boldsymbol{10ps}$.
- Choose one of the two methods shown in ***Figures 25*** and ***26*** to measure the timing margin.
- If the method shown in ***Figure 25*** is selected, complete the timing margin for each PVT corner as presented in the following table.



Parameter	Description
T_period_ideal	Ideal clock period
Tsetup_int	Intrinsic setup time (Characterize)
Thold_int	Intrinsic hold time (Characterize)
Tsetup_unc	Setup time uncertainty due to clock uncertainty (Calculate)
Thold_unc	Hold time uncertainty due to clock uncertainty (Calculate)
Tsetup_sim	Setup time of FF when simulating (Simulate)
Thold_sim	Hold time of FF when simulating (Simulate)
Tsetup_margin	Margin of setup time (must be > 0)
Thold_margin	Margin of hold time (must be > 0)



Tsetup_unc = Tclock_unc

Thold_unc = Tclock_unc

Tsetup_margin = Tsetup_sim – (Tsetup_unc + Tsetup_int)

Thold_margin = Thold_sim – (Thold_unc + Thold_int)

Figure 25. Proposal method to calculate timing margin

- Corner TTNN

Path	Tsetup_int	Tsetup_unc	Tsetup_sim	Tsetup_margin	Thold_int	Thold_unc	Thold_sim	Thold_margin
S0_ data0								
S0_ data1								
S1_ data0								
S1_ data1								

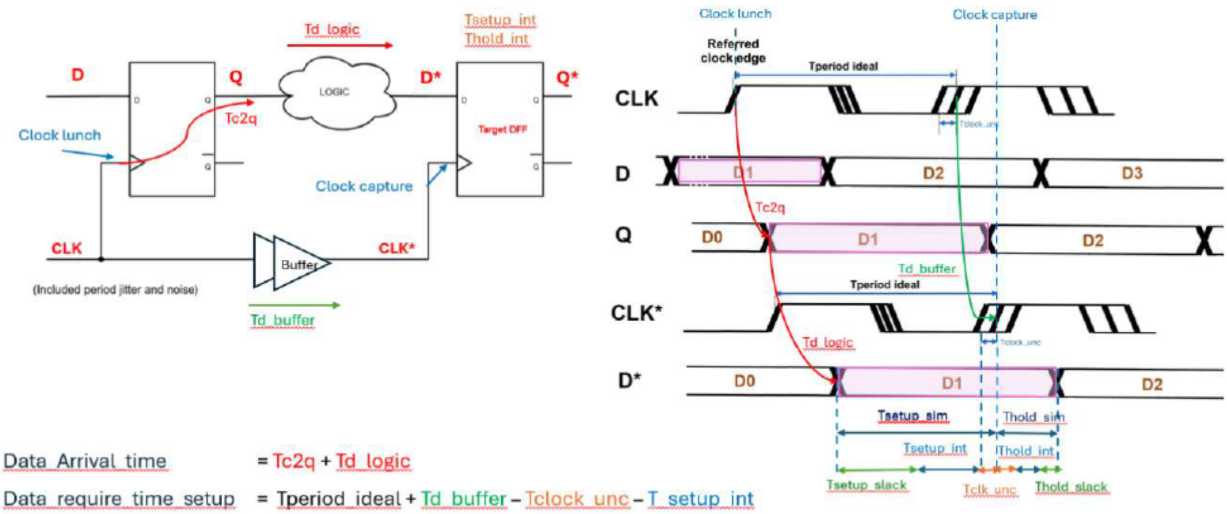
- Corner SSLL

Path	Tsetup_int	Tsetup_unc	Tsetup_sim	Tsetup_margin	Thold_int	Thold_unc	Thold_sim	Thold_margin
S0_ data0								
S0_ data1								
S1_ data0								
S1_ data1								

- Corner FFHH

Path	Tsetup_int	Tsetup_unc	Tsetup_sim	Tsetup_margin	Thold_int	Thold_unc	Thold_sim	Thold_margin
S0_ data0								
S0_ data1								
S1_ data0								
S1_ data1								

- Students may refer to the STA flow shown in ***Figure 26***, which provides the same timing margin (slack) results.



Data_Arrival_time = Tc2q + Td_logic

Data_require_time_setup = Tperiod_ideal + Td_buffer – Tclock_unc – T_setup_int

Data_require_time_hold = Td_buffer + Thold_int + Tclock_unc

Tsetup_slack = Data_require_time_setup – Data_Arrival_time

Thold_slack = Data_Arrival_time – Data_require_time_hold

Figure 26. Proposal method to calculate timing margin

- If the method shown in ***Figure 26*** (STA method) is selected, create and complete the corresponding timing margin table, follows to table from ***Figure 25*** method.